Clinical Practice Guideline: Lipid Profile & Cardiovascular Risk Management

1. Low-Density Lipoprotein (LDL) Cholesterol

LDL cholesterol is atherogenic (bad cholesterol). Optimal LDL is below 100 mg/dL. Near-optimal is 100 to 129 mg/dL, borderline high is 130 to 159 mg/dL, high is 160 to 189 mg/dL, and very high is 190 mg/dL or above. In patients with diabetes mellitus, target LDL should be strictly below 70 to 100 mg/dL.

2. High-Density Lipoprotein (HDL) & Triglycerides

HDL cholesterol is cardioprotective. Low HDL (< 40 mg/dL in men, < 50 mg/dL in women) increases heart disease risk; HDL >= 60 mg/dL is protective. Normal Triglycerides are < 150 mg/dL; 150-199 mg/dL is borderline high; 200-499 mg/dL is high; >= 500 mg/dL carries pancreatitis risk.

3. Combined Diabetic Dyslipidemia Protocol

When high LDL (>= 130 mg/dL) or high Triglycerides (>= 150 mg/dL) coexist with elevated HbA1c (>= 6.5%), cardiovascular risk is multiplied. Management requires eliminating trans fats, increasing soluble dietary fiber, and initiating statin therapy after checking baseline liver enzymes (ALT/AST).

Lipid Parameter	Optimal / Normal	Borderline High	High / Abnormal
Total Cholesterol	< 200 mg/dL	200 - 239 mg/dL	>= 240 mg/dL
LDL Cholesterol	< 100 mg/dL	130 - 159 mg/dL	>= 160 mg/dL
HDL Cholesterol	>= 60 mg/dL	40 - 59 mg/dL	< 40 mg/dL (Risk)
Triglycerides	< 150 mg/dL	150 - 199 mg/dL	>= 200 mg/dL